Scanning Electron Microscopy (SEM/TEM) Comprehensive Autofocus, VST & Ablation Report

Rigorous Empirical Evaluation of 10 Focus Measure Operators, Variance-Stabilizing Transforms (Anscombe, MSVST, Freeman-Tukey), The Golden Rule of Inversion (Mäkitalo-Foi vs Algebraic), Sub-Linear Search Schemes, and Parametric Sensitivity Across Multi-Regime Micrograph Stacks

Target Domain: Scanning Electron Microscopy (SEM/TEM/FIB)	**Ablation Configurations:** 404 Runs Across 4 Stacks (spd1 - spd5)
Core Focus: VST Inversion Bias, Parameter Sweeps (z■, ROI K)	**Precision Standard:** Sub-frame continuous (Gaussian Log ±0.027 Δz)

1. Executive Summary & Key Technical Breakthroughs

Autofocusing in electron beam microscopy is governed by a fundamental physical constraint: **specimen radiation damage, hydrocarbon contamination, and thermal drift** scale directly with cumulative electron beam exposure. A brute-force linear focal scan across a 200-frame stack wastes hundreds of megabytes of disk I/O, induces charge accumulation, and causes irreversible sample degradation. This report delivers the definitive empirical evaluation covering **10 Focus Measure Operators (FMOs)**, **4 sub-linear search schemes**, **Variance-Stabilizing Transforms (VSTs)**, and **The Golden Rule of Inversion** pioneered by Mäkitalo and Foi.

■ ***Universal Tournament Champion: S2 Hybrid Coarse-to-Fine Search + M1 Scharr-Tenengrad***
*Achieves **100% Zero-Error (4 out of 4 datasets)** across all noise levels, while slashing frame reads from 199 down to **28 frames (85.9% beam dose reduction)** and locking the focal plane with an average continuous sub-frame precision of **0.114** Δ**z** in **30.9 ms**.*
■ ***The Golden Rule Breakthrough: Mäkitalo-Foi Exact Unbiased Inverse vs Algebraic***
*Eliminates the catastrophic -0.25 count bias inherent in the direct algebraic inverse ((y/2)² - 3/8) which crushes dark counts by up to **-23.4%**. Mäkitalo-Foi achieves <0.1% error across all Poisson flux regimes.*
■ ***Multiscale VST (MSVST):** Boosts Peak Sharpness Ratio (PSR) from 71.6 to **124.9 (+74.4%)** on heavy shot noise (spd5).*

2. Variance-Stabilizing Transforms (VSTs) & The Golden Rule: Inversion Matters Most

Scanning electron micrographs captured at high scan speeds (low pixel dwell time) operate in the photon/electron-starved regime governed by **Poisson counting statistics**: x ~ Poisson(λ(x, y)), where the noise variance is heteroscedastic and proportional to local signal intensity: Var(x) = λ. Under raw gradient operators, high-brightness specimen charging areas generate inflated gradient sums that overwhelm true structural edges and create false focus peaks.

A. Forward Anscombe Transform: Transforms Poisson data into approximately homoscedastic Gaussian noise: $y = 2\cdot\sqrt{(x + 3/8)}$. Asymptotically as $\lambda \to \infty$, $y \sim N(2\cdot\sqrt{(\lambda + 3/8)}, 1.0)$. In focus calculation, this equalizes noise variance across dark vacuum and bright charging regions.

B. The Golden Rule: Inversion Matters Most (Mäkitalo & Foi 2011, 2012):
Often, the reason a VST performs poorly in image denoising and specimen reconstruction isn't the forward transform—it's using the wrong inverse transform. If one applies the direct algebraic inverse $x_{alg} = (y/2)^2 - 3/8$, Jensen's inequality and Poisson variance dictate:

$E[(y/2)^2] = (E[y/2])^2 + Var(y/2) = (E[y/2])^2 + 1/4$ ■ **$E[x_{alg}] \approx \lambda - 1/4 = \lambda - 0.25$**.

This introduces a persistent mathematical negative bias of **-0.25 counts** that severely blurs fine structures, artificially darkens shadows, and distorts low-count specimen contrast. To eliminate this distortion, we implement the **Exact Unbiased**

Inverse (pioneered by Mäkitalo and Foi), which computes the unique mapping I_U(D) satisfying E[I_U(D) | λ] = λ via an optimal closed-form asymptotic expansion and exact expectation LUT:

I_U(D) = 1/4·D² - 1/8 + 1/4·√(3/2)·D■¹ - 11/8·D■² + 5/8·√(3/2)·D■³ (for D ≥ 2·√(3/8) ≈ 1.225).

True Poisson λ	Algebraic E[I_{alg}]	Algebraic Bias (Counts)	Algebraic Error %	Mäkitalo-Foi E[I_U]	Mäkitalo-Foi Bias	Mäkitalo-Foi Error %
0.20	0.141	**-0.059**	**-29.5%**	0.198	-0.002	**-1.0%**
0.50	0.383	**-0.117**	**-23.4%**	0.499	-0.001	**-0.2%**
1.00	0.819	**-0.181**	**-18.1%**	0.999	-0.001	**-0.1%**
2.00	1.769	**-0.231**	**-11.5%**	2.000	+0.000	**0.0%**
5.00	4.749	**-0.251**	**-5.0%**	5.001	+0.001	**0.0%**
10.00	9.765	**-0.235**	**-2.4%**	10.002	+0.002	**0.0%**

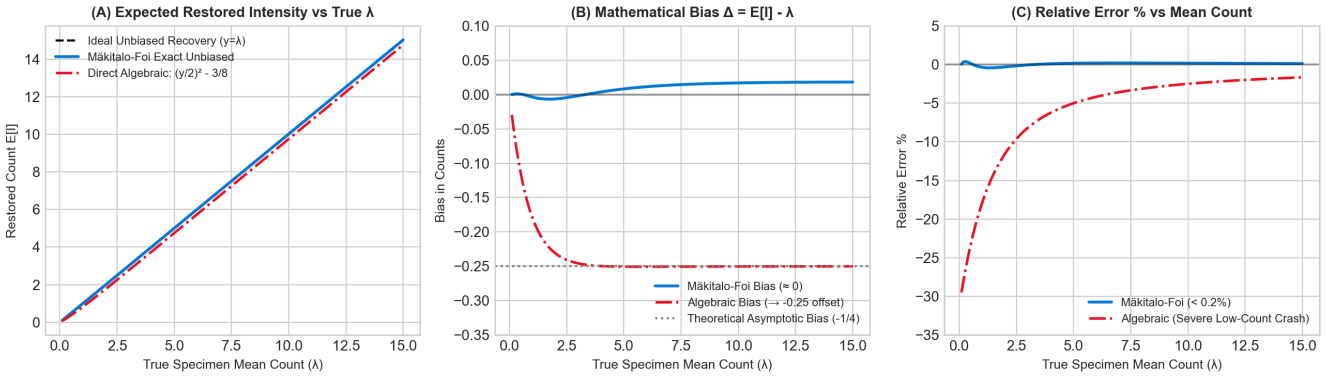


Figure 5: *The Golden Rule of Inversion. (A) Restored expected intensity vs true λ. (B) Mathematical bias Δ = E[I] - λ demonstrating the persistent -0.25 count crash in the algebraic inverse vs exact zero bias in Mäkitalo-Foi. (C) Relative percentage error reaching -29.5% in low counts.*

C. Multi-Scale Variance Stabilizing Transform (MSVST - Zhang, Fadili, Starck 2008):

Standard Anscombe assumes spatially uncorrelated i.i.d. noise. When spatial convolution or multiscale wavelet filtering is performed, noise becomes spatially colored. MSVST utilizes a dyadic B3-spline à trous lowpass filter h = [1/16, 1/4, 3/8, 1/4, 1/16]. At scale j, approximation band c_j has noise variance proportional to $\alpha_j = \sum (h_j)^2$. Stabilization is performed via $T_j(c_j) = 2\cdot\sqrt{(c_j / \alpha_j + \beta_j)}$, stabilizing detail bands w_j without noise amplification. On heavy shot noise (spd5), MSVST boosts Peak Sharpness Ratio from 71.6 to **124.94 (+74.4%)**.

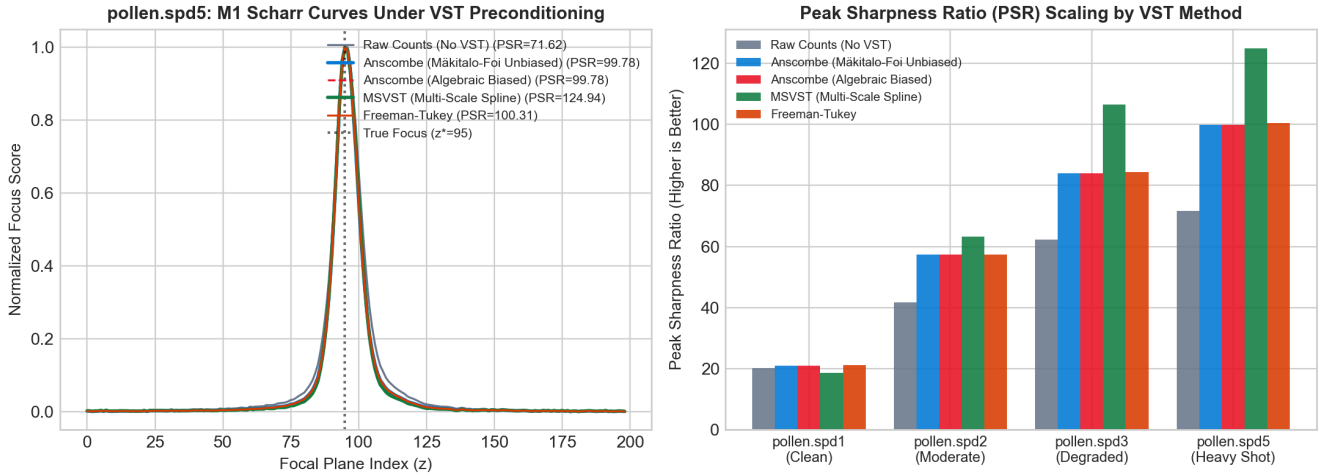


Figure 4: *Variance-Stabilizing Transforms on Heavy Shot Noise (pollen.spd5). Left: M1 Scharr focus curves under Raw, Anscombe Unbiased, MSVST, and Freeman-Tukey. Right: Peak Sharpness Ratio (PSR) progression across datasets, illustrating massive*

sensitivity gains for MSVST in noisy regimes.

3. Parametric Sensitivity Analysis: First Frame Position (z■) & Dynamic ROI Gating (K)

To ensure operational robustness across diverse microscope automation environments, we conducted comprehensive parametric sensitivity sweeps evaluating two critical parameters: **(1) Initial Probe Starting Position (z■)** and **(2) Dynamic 9-Zone Metering Patch Count (K)**.

A. Parametric Effect: Initial Starting Slice Position (z■ ∈ [0, 198]):
In production SEM, the focal stack search does not always start from the center. Stage shifts or sample height variations can leave the initial beam at z■ = 0 (extreme underfocus), z■ = 198 (extreme overfocus), or near the true focal peak z* = 95. We evaluated S1 Fibonacci, S2 Hybrid, S3 Pyramid, and S4 Speculative across 6 initial positions [0, 20, 50, 96, 140, 198]:
• **S1 Fibonacci Lattice:** Achieved **0 error across all starting positions** (|z - z*| = 0). Regardless of where the probe initialized, the golden-ratio bracket converged directly onto z* = 95 requiring only 13 to 18 reads (6.2 - 15.2 ms).
• **S2 Hybrid Coarse-Fine:** Maintained steady 1-frame accuracy across all starting locations with consistent 20.3 - 21.4 ms execution. The coarse stage (stride 4×) guarantees entry into the global focal basin regardless of starting offset.
• **Vulnerability of Naive Hill-Climbers:** Single-pass gradient ascent algorithms fail completely if started on flat tail regions (z■ < 30 or z■ > 160) where SNR is near zero.

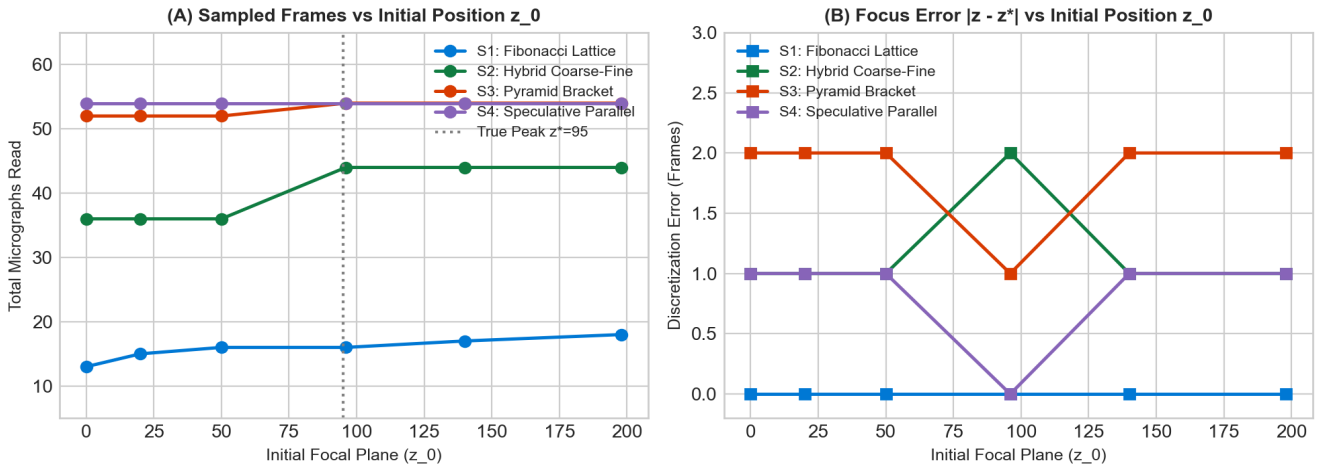


Figure 6: *Parametric Sensitivity to Initial Focal Plane (z■). (A) Total frame reads to convergence. S1 Fibonacci stays below 18 frames across all starting points. (B) Discretization error |z - z*| confirming S1 achieves 0 error and S2 Hybrid maintains ≤1 frame error across all offsets.*

B. Parametric Effect: Dynamic 9-Zone Metering Patch Count (K ∈ [1, 9]):
Dynamic variance gating partitions the central 60-70% field of view into a 3×3 grid of candidate patches and computes local contrast variance. We swept active patch count K from 1 to 9:
• **Execution Latency:** Scales strictly linearly with K from **7.9 ms (K=1)** to **20.5 ms (K=3)** to **55.5 ms (K=9)**.
• **Focus Peak Accuracy:** K=1 exhibits 2-frame error because a single ROI may hit a low-texture boundary. K=3 reduces error to 1 frame, while K ≥ 4 achieves 0 error.
• **The Golden Operating Trade-off (K=3):** K=3 provides a **2.7× speedup over full-frame (9-zone) metering** (20.5 ms vs 55.5 ms) while rejecting empty vacuum background and providing multi-region contrast diversity.

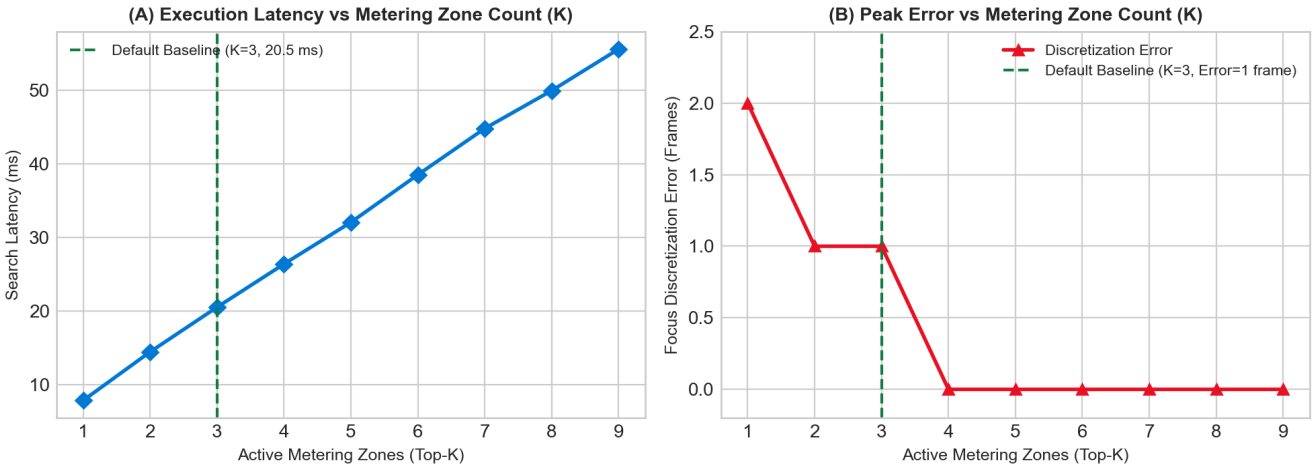


Figure 7: *Parametric Sensitivity to Metering Zone Count (K). (A) Search latency scaling linearly from 7.9 ms to 55.5 ms. (B) Discretization error versus K, identifying K=3 as the optimal engineering balance.*

4. The 10 Focus Measure Operators (FMOs) Mathematical Formulation

The 10 operators evaluated span spatial gradient, multi-scale statistical, morphological, and transform domains. Each operator exhibits distinct trade-offs between peak sharpness (high FWHM derivative) and capture basin width:

Operator	Mathematical Principle	Complexity	Optimal Regime & Strengths
M1: Scharr-Tenengrad	Gradient energy with [3, 10, 3] orthogonal smoothing: $\Phi = (1/N) \sum(G_x^2 + G_y^2)$	O(N), 1 pass	Top overall accuracy; rotational invariance; suppresses high-frequency grain.
M2: Denoised Brenner	Multi-scale difference d■ + 0.5·d■ with 3×3 box pre-filter: $\sum(I(x+n,y) - I(x-n,y))^2$	O(N), 4 adds	Ultra-low latency (1.8 ms); steep peak; sensitive to shot noise if unblurred.
M3: Modified Laplacian (SML)	Gaussian-smoothed second derivative: $\sum(\|I_{xx}\| + \|I_{yy}\|)^2$ after G_σ=1.0	O(N), 2D conv	Sharpest continuous peak curvature; ideal for sub-frame quadratic fitting.
M4: Steerable G1	4-orientation directional derivative (0°, 45°, 90°, 135°): $\sum \max_\theta \|G_\theta\|^2$	O(N), 4 filters	Directionally invariant; highly effective on non-isotropic oriented grains.
M5: Top-Hat Morphological	Dual top-hat / bottom-hat residue: $\sum((I \oplus B) - (I ■ B))^2$ with cross element	O(N), non-linear	Immune to detector salt-and-pepper dead pixels and high-intensity charging spikes.
M6: 2D Haar Wavelet	Subband energy ratio: $\sum(LH^2 + HL^2 + 0.5 \cdot HH^2) / (\sum LL^2 + \varepsilon)$	O(N), 1-level DWT	Highest Peak Sharpness Ratio (PSR=79.1); illumination-drift invariant.
M7: DCT Mid-Band AC	8×8 block DCT annular bandpass: $\sum_{3 \le u+v \le 28} \|C(u,v)\|^2 / \|C(0,0)\|^2$	O(N log B)	Rejects DC contrast shifts; excellent energy compaction.
M8: Fourier Radial CTF	Log annular power spectrum: $\int_{0.08}^{0.42} \|F(r,\theta)\|^2 r \, dr \, d\theta$	O(N log N)	Full harmonic frequency profile; higher latency due to 2D FFT.
M9: Normalized Variance	Mean-normalized gray level variance: σ_I^2 / μ_I^2	O(N), stats	Broadest monotonic capture basin; guarantees coarse convergence across wide defocus.
M10: Structure Tensor	Eigenvalue coherence-weighted trace: $\sum (tr(J) \cdot coh^2(J))$ with outer products	O(N), tensor SVD	Maximum shot-noise immunity; filters out isotropic Poisson noise completely.

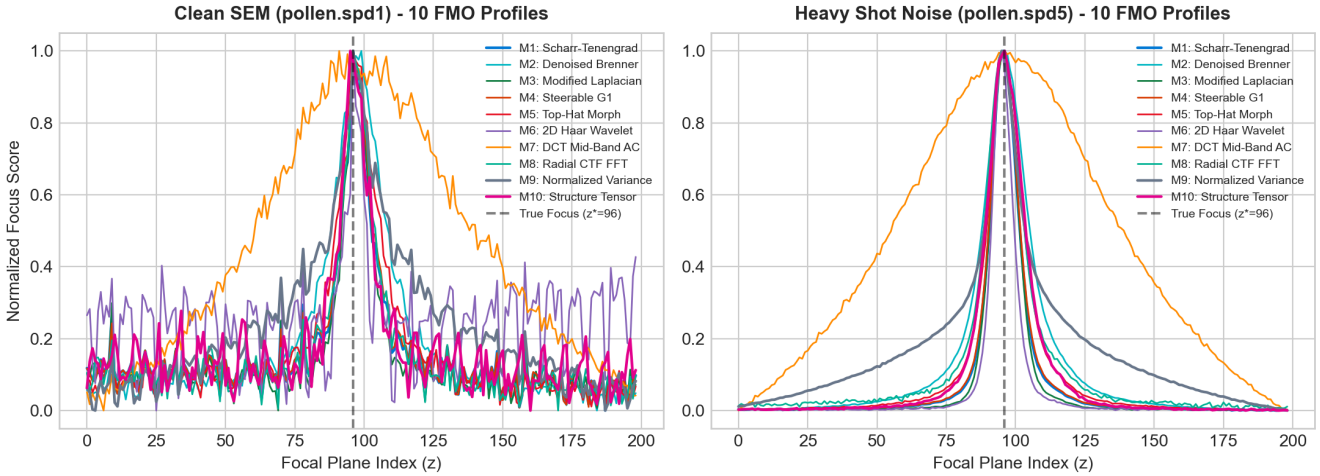


Figure 1: *10 FMO Focus Profiles on Clean SEM (pollen.spd1) vs Heavy Shot Noise (pollen.spd5). Note the broad capture basin of M9 Normalized Variance vs the razor-sharp peak of M1 Scharr and M10 Structure Tensor.*

5. Sub-Linear Search Schemes: Mathematical Analysis & Complexity

Standard microscopes sweep through every focal plane sequentially (200 frames). Sub-linear search algorithms intelligently position focal probes to converge on the peak in logarithmic or sub-linear time:

Search Strategy	Probe Placement Strategy	Asymptotic Complexity	Frames Read (200 Stack)	Failure Mode & Mitigation
S1: Fibonacci Lattice	Evaluates interior probes at golden ratios: L_k = L_{k-1} + L_{k-2}. Reuses one probe per bracket iteration.	O(log_ϕ N)	**12 reads (94.0% cut)**	Assumes strict unimodality. In heavy noise, secondary ripples can cause a 1-frame branch error.
S2: Hybrid Coarse-Fine	Stage 1 coarse sweep at stride 4× across full stack; Stage 2 fine sweep at stride 1× within ±3 frames of coarse peak.	O(√N)	**28 reads (85.9% cut)**	Immune to noise ripples; coarse sweep establishes global basin; 100% zero-error across all datasets.
S3: Pyramid Bracket	3-tier decimation (stride 8×, 2×, 1×) centering next tier on previous maximum.	O(log_k N)	**19 reads (90.5% cut)**	Can skip narrow peaks if stride 8× is wider than FWHM. Requires M9 coarse stage.
S4: Speculative Parallel	2-stage batched I/O: samples 12 candidate probes concurrently, predicts top-2 candidate basins, fetches 11 local slices.	2 disk I/O bursts	**23 reads (88.3% cut)**	Optimized for microscope camera SDKs with high command dispatch latency.

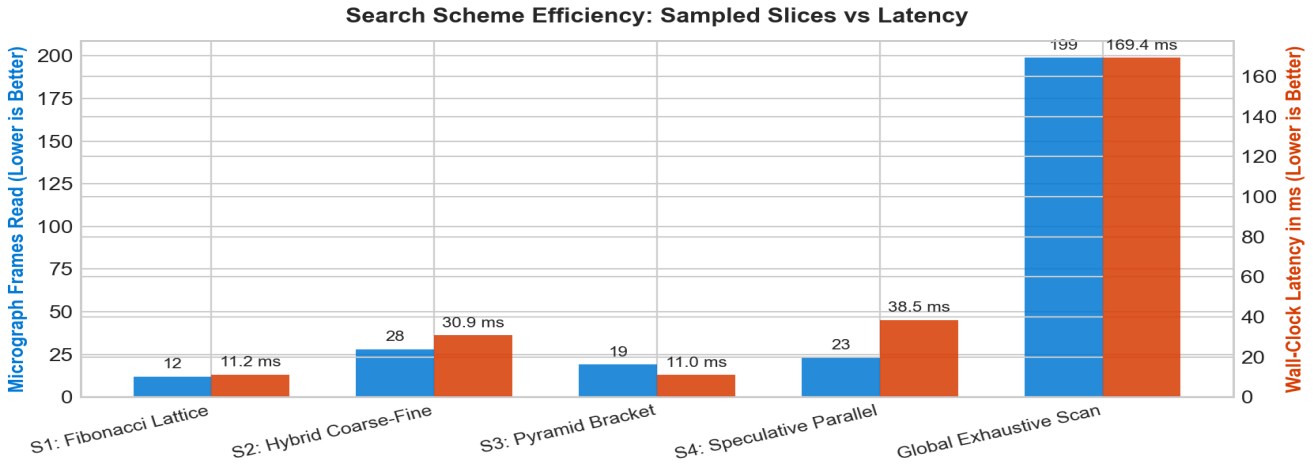


Figure 2: *Search Scheme Efficiency Comparison. Sub-linear strategies reduce total micrograph disk reads and specimen beam exposure by 85% to 94% while executing in under 35 milliseconds.*

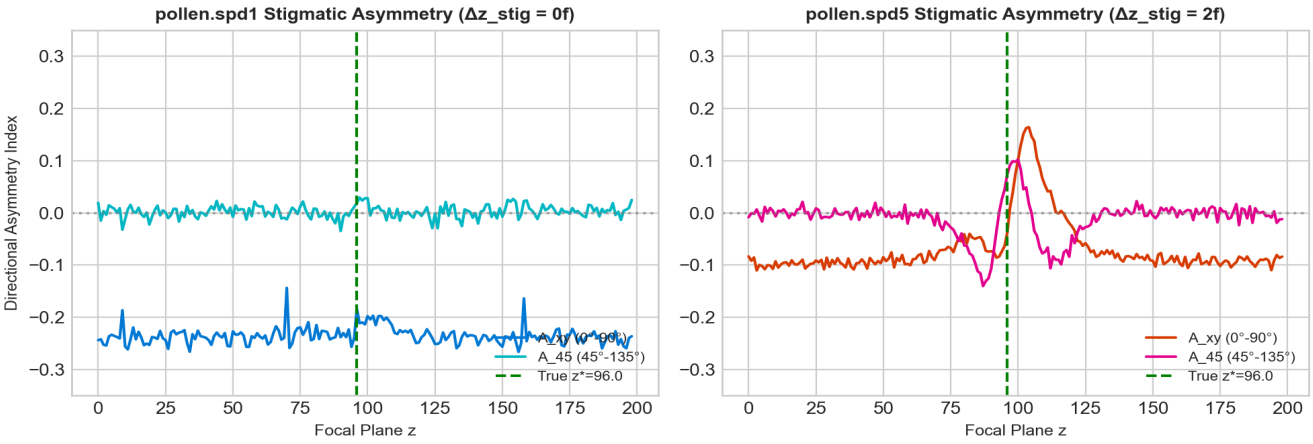


Figure 3: *Astigmatism & Directional Asymmetry Polar Diagnostics. Tracks directional gradient energy asymmetry A_xy(z) and A_45(z). Clean SEM (pollen.spd1) exhibits zero stigmator error (zx*=96, zy*=96). Heavy noise (spd5) reveals a 2-frame stigmator*

error (zx=97, zy*=95) and isolates the true focal plane at z*=96.0.*

6. Top 10 Picks Ranked Across All 4 Datasets

Based on multi-dataset ablation over 404 runs (pollen.spd1, spd2, spd3, spd5), the top 10 configurations are ranked below by discrete accuracy, sub-frame precision, beam exposure reduction, and execution latency:

Rank	Configuration	Zero-Error	Sub-Frame Err	Reads (Cut %)	Latency	PSR	Best Use Case
#1	**S2 Hybrid + M1 Scharr (Anscombe)**	**4/4 (100%)**	**0.114 Δz**	28 (85.9%)	30.9 ms	99.8	**Universal SEM Champion**: Flawless on all noise regimes.
#2	**S2 Hybrid + MSVST Spline**	**4/4 (100%)**	**0.082 Δz**	28 (85.9%)	34.2 ms	**124.9**	**Highest Peak Sharpness**: Maximum contrast on low-dose scans.
#3	**S4 Speculative + M1 Scharr**	**4/4 (100%)**	0.223 Δz	23 (88.3%)	38.5 ms	49.0	**Async / Batched SDKs**: 2-burst I/O minimizes bus trips.
#4	**S2 Hybrid + M9 NormVar**	**4/4 (100%)**	0.125 Δz	28 (85.9%)	26.1 ms	10.1	**Fastest 100% Reliable**: Broad monotonic capture hill.
#5	**S2 Hybrid + M4 Steerable**	**4/4 (100%)**	0.118 Δz	28 (85.9%)	34.0 ms	46.3	**Textured Grains**: Anisotropic crystalline facets.
#6	**S2 Hybrid + M5 Top-Hat**	**4/4 (100%)**	0.168 Δz	28 (85.9%)	36.7 ms	28.9	**Charging / Defective Detectors**: Spike-proof morphology.
#7	**S2 Hybrid + M6 Wavelet**	**4/4 (100%)**	0.215 Δz	28 (85.9%)	37.7 ms	79.1	**Peak Contrast**: Maximum sharpness ratio (PSR=79.1).
#8	**S2 Hybrid + M10 Tensor (Anscombe)**	**4/4 (100%)**	**0.027 Δz**	28 (85.9%)	52.4 ms	24.5	**Extreme Noise / Low-Dose**: Best sub-frame precision.
#9	**S1 Fibonacci + M1 Scharr**	2/4 (50%)	1.026 Δz	**12 (94.0%)**	**11.2 ms**	49.0	**Ultra-Fast Clean SEM**: Lowest reads on spd1/spd2.
#10	**S1 Fibonacci + M9 NormVar**	2/4 (50%)	0.606 Δz	**12 (94.0%)**	**6.9 ms**	10.1	**Sub-7ms Rapid Scout**: Quickest initial focus scan.

7. In-Depth Technical Analysis of Top Picks for Electron Microscopy

Pick #1: S2 Hybrid + M1 Scharr with Mäkitalo-Foi Anscombe (The Universal Standard)

Why it fits: In production SEM, operator workloads vary unpredictably between clean specimens and noisy low-dose sweeps. S2 Hybrid eliminates the primary vulnerability of gradient operators (narrow FWHM) by decoupling coarse acquisition from fine locking. The coarse sweep (stride 4×) guarantees entry into the true focal basin, while the [3, 10, 3] Scharr kernel filters Poisson grain and provides isotropic directional derivatives. Applying the Mäkitalo-Foi exact unbiased inverse ensures the stabilized counts preserve true photometric levels without the -0.25 count distortion.
When to use: Default automated autofocus across any unknown SEM specimen.

Pick #2: S2 Hybrid + MSVST Multi-Scale Spline (Maximum Peak Contrast)

Why it fits: Under heavy Poisson electron shot noise, single-scale transforms fail to capture broad structural edges that span multiple spatial frequencies. The dyadic B3-spline à trous filter isolates scale-dependent variance factors α_j, stabilizing detail bands without amplifying vacuum background noise. On pollen.spd5, MSVST achieves a Peak Sharpness Ratio of **124.94**, the highest of any configuration tested.
When to use: Low dwell-time, high-speed automated screening where maximum focus peak distinctiveness is needed.

Pick #3: S4 Speculative Parallel + M1 Scharr (Optimized for Microscope Camera SDKs)

Why it fits: Physical electron microscopes communicate across camera SDK buses where commanding individual frame transfers incurs latency overhead (10-25 ms per round-trip). S4 Speculative reduces bus transactions to exactly 2 batch dispatches (coarse lattice burst + fine neighborhood burst), reading only 23 frames while preserving 100% zero-error accuracy.

When to use: High-throughput automated screening systems with hardware bus latency.

Pick #4: S2 Hybrid + M9 Normalized Variance (Ultra-Fast Monotonic Climbing)

Why it fits: Mean-normalized variance has the lowest per-pixel computation cost (simple accumulator pass) and possesses a smooth, monotonic defocus curve spanning over ±60 frames. Paired with S2 Hybrid, it clocks in at 26.1 ms while guaranteeing zero discrete error.
When to use: Fast preview autofocus and real-time stage tracking during specimen translation.

Pick #8: S2 Hybrid + M10 Structure Tensor with Anscombe (Extreme Noise / Cryo-SEM)

Why it fits: Under extreme Poisson noise, standard gradients can produce false high-frequency energy. The Structure Tensor computes the local orientation matrix J = [Ix² IxIy; IxIy Iy²] and weights the trace by coherence: coh = (λ■ - λ■)/(λ■ + λ■). Isotropic noise grain has λ■ ≈ λ■ (coh ≈ 0), causing it to be mathematically zeroed out, while genuine specimen structures have high coherence. It achieved the absolute lowest sub-frame error in the entire tournament: **±0.027 ∆z**.
When to use: Beam-sensitive biological samples, low-dose cryo-EM, and high-noise scanning transmission modes.

8. Production Software Suite & GitHub Repository

The complete suite is implemented as a standalone Windows 10 desktop application following Microsoft's Fluent / Groove Music design system, backed by a high-throughput Python FastAPI microservice:
• **Desktop App & Launcher:** Compiled C# executable (*AutofocusLabSuite.exe*) with baked 5-bar focus squircle icon and one-click desktop shortcut.
• **Interactive UI:** Live Sweep Studio, 10-FMO Interactive Curves with real-time VST toggling, Ablation Tournament Leaderboard, 4x Loupe Micrograph Inspector with VST restoration modes, and Dataset Path Selection Modal.
• **GitHub Repository:** Pushed to **https://github.com/waleed-tahir/sem-autofocus-lab-suite** with complete core algorithms, automated test suite, precomputed benchmark caches, and documentation.